

LAITH NAYAL

Innopolis, Republic of Tatarstan, Russia

+7 989 114 9847 ◇ nayallaith2002@gmail.com ◇ l.nayal@innopolis.university ◇ [GitHub](#) ◇ [Google Scholar](#) ◇ [ResearchGate](#)

EDUCATION

Innopolis University

September 2022 - July 2026

Bachelor of Science, Computer Science, 4th Year. Innopolis, Republic of Tatarstan, Russian Federation

- Full scholarship funded by Innopolis University

RESEARCH INTERESTS

- Generative Modeling, Computer Vision, Multi-Modality, Quantum Computing.

SCIENTIFIC PAPERS

- **Laith Nayal, Mahmoud Moustat, Bader Rasheed.** “Robustness through Lipschitz-Guided Stochastic Depth in Neural Networks.” *arXiv Preprint*.
- **Laith Nayal, Hadi Salloum, Ahmad Taha, Yaroslav Kholodov, Alexander Gasnikov.** “Training-Free Out-of-Distribution Segmentation with Foundation Models.” *Proceedings of the International Conference of Computational Optimization (ICOMP 2025)*.
- **Hadi Salloum, Osama Orabi, Yaroslav Kholodov, Ahmad Taha, Marko Pezer, Hamza Shafee Aldaghstany, Alaa Aldin Hajjar, Laith Nayal, Sergey Savin, Mirko Farina.** “Q GPT: A New Direction in Quantum AI Integrating Large Language Models with Quantum Optimization for Accelerated Industrial Problem Solving.” *Submitted to Scientific Reports (Nature), 2026*.
- **Hadi Salloum, Laith Nayal, Manuel Mazzara.** “Evaluating the Advantage 2 Quantum Annealer Prototype: A Comparative Evaluation with Advantage 1 and Hybrid Solver and Classical Restricted Boltzmann Machines on MNIST Classification.” *Proceedings of the 13th International Conference on Mathematical Modeling in Physical Sciences, 2024*.

EXPERIENCE

Multimodal Generative AI Research in Industry, a joint laboratory of Fusion Brain at AIRI and Innopolis University

August 2025 – Present

Researcher

Innopolis, Republic of Tatarstan, Russian Federation

- Working on **multilingual text rendering using diffusion models and training acceleration techniques for diffusion models**, under the supervision of Prof. Andrey Kuznetsov and Prof. Aibek Alanov.

Quantum Computing Lab, Innopolis University

March 2025 – August 2025

Lab Assistant

Innopolis, Republic of Tatarstan, Russian Federation

- Contributing to the development of a high-quality research paper exploring novel methods for modeling uncertainty in deep learning systems under the supervision of Professor Yaroslav Kholodov

Q Deep

September 2024 – May 2025

AI Research Engineer

Innopolis, Republic of Tatarstan, Russian Federation

- Conducting research and contributing to the development of high-quality papers at the intersection of quantum computing and machine learning.

HUB21 Learning UK Limited

September 2023- Present

Python and Algorithmic Thinking Tutor at Hub21.

London, United Kingdom

- Hub21 is a learning platform with the mission to help students find their passion and achieve their dreams in the technological fields.

Competitive programmer

January 2021 - Present

Contestant

Online

- I actively engage in competitive programming and participate in various online and onsite competitions, including Meta Hacker Cup, ACM-ICPC, Codeforces, and Atcoder.

ACHIEVEMENTS

Full Scholarship At Innopolis University

September 2022 - 2026

Recipient

Innopolis, Republic of Tatarstan, Russian Federation

- A merit-based award for highly achieving students to pursue a bachelor degree in Computer Science.

Research Award

August 2025

Winner

Innopolis, Republic of Tatarstan, Russian Federation

- A monthly scholarship for scientific achievements, presented by the Rector of Innopolis University, Professor Alexander Gasnikov.

Student Of The Year

2025, 2026

Winner

Innopolis, Republic of Tatarstan, Russian Federation

- I received the Student of the Year award for the years 2025 and 2026 for excellent scientific achievements.

The ICPC Quarterfinals in Saratov

November 2025

Contestant

Saratov, Russian Federation

- Honorable Mention Certificate.

The ICPC Southern and Volga Russia Qualification

October 2025

Contestant

Innopolis, Republic of Tatarstan, Russian Federation

- Second Award Certificate.

Winner of a Sport Programming Bootcamp

23-27, April, 2024

Bronze Medalist

MIPT, Moscow, Russian Federation

- The event is organized by the School of Applied Mathematics and Informatics of the Moscow Institute of Physics and Technology with the support of VK Education and SberTech.
My team and I took 3rd place in division A.

Meta hacker cup

2024

Contesant

Online

- Participated in Two rounds, Received a Certificate of Recognition.

ICPC Online Challenge powered by Huawei

November 2023

Contestant

Online

- Received a Honorable Mention.

SKILLS

- **Technical:** Python, C++, PyTorch, CUDA, Multimodal Diffusion Transformers (MMDiT), Diffusion Models, Vision-Language Models, Parameter-Efficient Fine-Tuning (PEFT, LoRA), Vision Foundation models, Representation Learning, Distributed Training (DDP), Multi-GPU/Multi-Node Training, High-Performance Computing (Slurm), Mixed Precision Training (BF16/FP16), Computer Vision, OCR-based Evaluation, Data Analysis.
- **Languages:** Arabic (native), English (fluent), French (Elementary), Russian(Elementary).

PROJECTS

Sudoku Classification Project CNN vs Generative Models

November 2024

Machine Learning

Innopolis, Republic of Tatarstan, Russian Federation

- Explored the classification of Sudoku and non-Sudoku images using three neural network architectures: Auto-Encoder (AE), Variational Auto-Encoder (VAE), and Convolutional Neural Networks (CNN).
- Conducted experiments to evaluate and compare the performance of each architecture on a dataset of 3x3 sudoku and non-sudoku images based on MNIST digits.
- Further details, including the dataset, implementation, and results, can be found in the provided GitHub repository.

Movies Recommendation System Using BERT

July 2024

Machine Learning

Innopolis, Republic of Tatarstan, Russian Federation

- Developed a transformer-based movie recommendation system using BERT embeddings to suggest movies and series from major streaming platforms. The project involved data preprocessing, embedding generation, and creating an API using FastAPI, as well as a user-friendly interface with Streamlit for easy interaction.

Stamp Recognition Project

July 2023

Machine Learning

Innopolis, Republic of Tatarstan, Russian Federation

- As part of a team, I contributed to developing and implementing a stamp detection, classification, and comparison solution using Machine Learning and Computer Vision techniques. The project involved executing data augmentation, applying machine learning for stamp classification and embedding extraction, and deploying the finalized solution with an API for practical use..

EXTRACURRICULAR ACTIVITIES

Inno Typeracer Club

September 2023 - Present

Leader & Founder

Innopolis, Republic of Tatarstan, Russian Federation

- The club is dedicated to organizing and hosting Typing competitions.